SCIF30005 Week 16 Workshop

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Outline

This week's workshop:

- data visualisation and report writing

Up next:

- mini project – instructions released today, video also available under this week's page on Blackboard

Outline

1. Mentimeter
2. Reports
3. Workshop
4. Summary

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What Makes a Good Report?

→ https://www.menti.com

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Scientific Reports

- should be telling a *story*, including the context, the gap your work is filling, how you did that, what you found
- usually involves answering a question – think about what that is, what data you need to answer, what your hypothesis might be
- not the same as a technical report or manual, though may be some overlap
- should be standalone – don't assume the reader has attended the workshops, read the instructions etc.
- think about the audience – could be a classmate, or you before you started the project
- there are various resources available, e.g. study skills page `https://www.bristol.ac.uk/students/your-studies/study-support/study-skills/`

Structure

- typical structure – abstract, introduction, methods, results (and discussion), conclusion
- may wish to vary section headings and/or add subsections, but the general structure should be the same
- don't forget to link between sections, e.g. phrases like "having introduced the image blurring model, we now describe the paralellisation approach"
- formatting and overall presentation:
  - do want basic things like good, clear English, page numbers, sensible font etc.
  - but doesn't matter if it's e.g. LaTeX or Word – different schools may have different guidelines for this
  - should look coherent, *consistent* and professional

Abstract

- should be short, often follows similar structure to the report as a whole
- i.e. needs to include some background, the key aims (i.e. what is your work trying to solve), the main results, and the implications
- idea is to capture reader's attention/help them figure out if your work is relevant to them
- various resources available, e.g. `https://communities.springernature.com/posts/how-to-write-an-abstract`

Introduction

- needs to answer various questions:
  - what is the background?
  - what is the motivation?
  - what has already been done (usually in the literature, but could be more general)?
  - what is your work going to add?
- needs to draw the reader in – should start broad and end with your specific problem
- could be mostly focused on a scientific problem, but may also be technical, e.g. might mention importance of HPC
- ends with an overview/outline of the report
- can be fairly short – typically no more than a page for a 5-6 page report

Methods

- needs to contain a description of what you did, including algorithms, key parameters etc.
- should not be too focused on technical details – not the same as a technical report or manual
- should be possible to reproduce your work from the description, but potentially using a different language
- $\rightarrow$ although you may wish to mention the code was written in C++, sharing specific C++ commands is not usually helpful
- an exception might be if you've implemented a new feature, but this should not be a line by line code breakdown
- similar with MPI: don't copy/paste the MPI command, do explain how the communication approach works
- code snippets are also not usually needed, but schematics/diagrams can be helpful
- analogy with experiment – don't need to describe standard glassware or experimental setups, unless it was something unique and custom you came up with

Methods: Bad Example

"The code was parallelised using MPI by defining a function to return values for the variables i0 and i1, which were used for the loop over the grid points. By looping from i0 to i1 on each task instead of 0 to Ni, the work was divided up between tasks. The vector new_image was then updated following the rules of the image blurring algorithm. Next, the grid was communicated using the call MPI_Allreduce(new_image[i][j].data(), old_image[i][j].data(), 3, MPI_INTEGER, MPI_SUM, MPI_COMM_WORLD);, so that old_image was now ready for the next iteration."

- true, but not helpful
- not clear how i0 and i1 were calculated – what kind of distribution, was the data evenly divided?
- variable names are irrelevant
- too specific to C++ – I might want to implement this in fortran, so telling me the C++ MPI syntax is just confusing
- …

Methods: Better Example

"The code was parallelised using MPI. First, the work was divided among tasks, so that each task was responsible for updating a subset of columns. To ensure load balancing, this was done as evenly as possible. Each task then updated its own columns, using the rules of the image blurring algorithm. Grid points associated with the remaining columns were set to zero, so that a reduction-based strategy could be used. The updated grid was then communicated using the collective MPI call MPI_Allreduce, where MPI_Allreduce was used instead of MPI_Reduce to make the updated grid available on all tasks, ready for the next iteration."

- less specific to C++, no needless variable names
- still specifies the exact MPI call used, as this tells us useful information
- explains not just what was done but *why*
- would be easier to reproduce in another programming language
- …

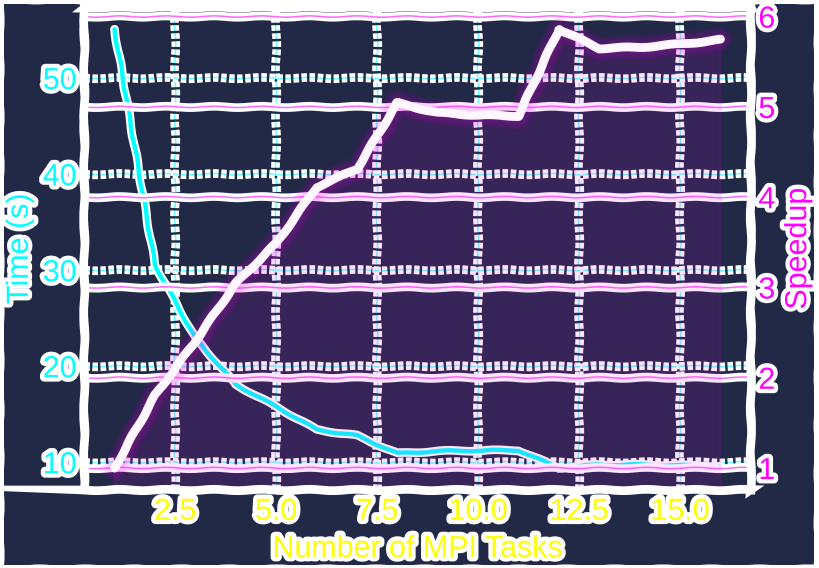
Results

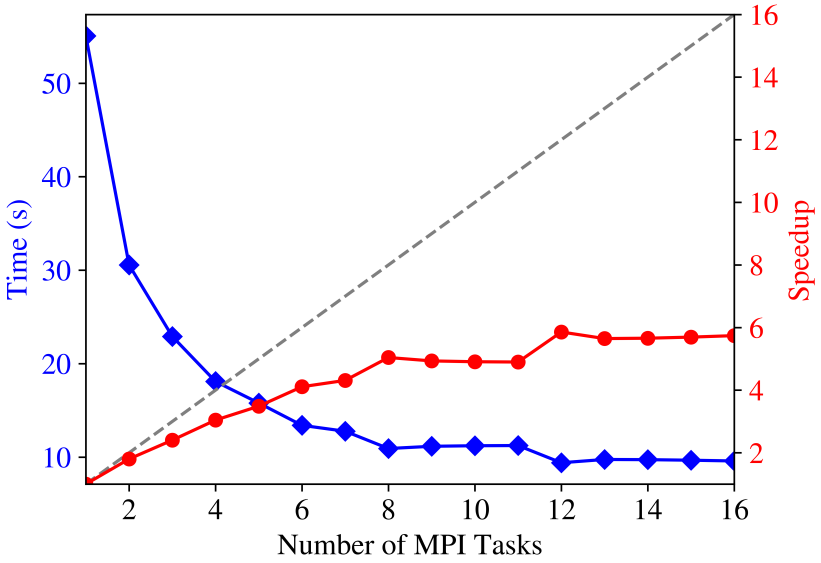
- don't just state what the results show, also *explain* them
- make sure you have enough data to backup your conclusions
- *all* figures and tables should be referred to in the text – if you don't refer to it, then why are you including it?
- captions should be self-contained (no titles), so e.g. define acronyms or key parameters, or if needed refer explicitly to the text
- may sometimes be appropriate to give values in the text, but make sure important results don't get lost
- don't give results to too many significant figures – how many are actually meaningful?
- *think* about what you expect to see – this will also help you spot mistakes
- if it's not clear what results mean, or results are unexpected, it's ok to say that
- but try and be constructive – what more could you do in future to help explain these results
- if relevant, put the results in context with previous work

Figures: General Tips

- first think about what information you are trying to convey
- think about the best type of plot, whether to use a logscale etc. – see TB1
- make sure the font size is large enough, label all axes
- think about the colour scheme – consider using a tool to check if colourblind-friendly, but also use different linestyles/markers etc.
- don't overcomplicate things – simpler is often better
- if comparing data, then side by side or combined plots may be better – having to flip between pages can mean the message gets lost
- think about the number of figures – if given a page limit, this should help guide you
- a table may be more appropriate, e.g. for precise reference values or presenting data which isn't clear in a graph
- but generally don't want a figure and table for the same data
- if fitting a line – what is the *meaning* behind it?

Figures: Bad *vs.* Better Example





- markers *vs.* just lines
- colour scheme
- gridlines
- tic locations
- ‘ideal’ speedup
- …

Used xkcd and cyberpunk packages – see e.g. `https://www.kaggle.com/code/parulpandey/10-advanced-plots-in-matplotlib`

Conclusion and References

- conclusion:
  - summarise what you did and what you discovered
  - include ideas for future work, e.g. how would you extend the model, what further calculations would you do given more time etc.
  - are there any wider implications, but be specific, e.g. MPI is useful for parallelising code is not helpful, but saying how you might use lessons learned to parallelise other cellular automata might be
  - can be fairly short, e.g. 1-2 paragraphs
- references:
  - may not be (m)any for this kind of report
  - don't forget to reference any figures you've reused, including e.g. course materials
  - be *consistent* with formatting

Use of AI

- the mini project is AI category 3: selective
- this means things the following are permitted:
  - help with spelling and grammar, advice/feedback on drafts
  - help with debugging, suggestions of ways to improve your code
- but these are still *not* allowed:
  - using AI to write or re-write sections of text
  - using AI to write or re-write sections of code
- make sure you're familiar with university guidelines, and if in doubt ask me
- you should also include a citation of technologies used, covering software usage including but not limited to AI

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Workshop: Writing

- taking the image blurring model, with the MPI send/receive implementation
- write an outline for the report, i.e. section headings, bullet points for what you would include
- write up the general rules and the MPI send/receive implementation
- swap with a friend/ask the GTAs or me for feedback

Workshop: Data Visualisation

- I've provided some timing data or use your own
- make the best and worst possible graph you can
- matplotlib inspiration gallery: `https://matplotlib.org/3.1.1/gallery/index.html`
- list of top ten worst graphs `https://www.biostat.wisc.edu/~kbroman/topten_worstgraphs/` – how would you plot these differently?

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Summary

In this workshop:

- we looked at what goes into a scientific report
- we looked at some of the features of a good graph